The Krawtchouk Matching Matrix: A Discrete Analogue of the Hermite Matching Matrix

Robin Langer and Claude Code

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Abstract

The Hermite matching matrix D_{mk} records the change of basis between the Hermite functions (eigenfunctions of the Fourier transform) and the Hermite polynomials. Its entries are integers that count signed bipartite matchings, with generating function $\sqrt{2\pi/3}\,\exp(-(s^2 - 4su + u^2)/3)$.

We construct the discrete analogue: the *Krawtchouk matching matrix*, which records the interaction between the Krawtchouk polynomials (eigenvectors of the binary Fourier transform) and a binomial weight at parameter q. Its generating function is the n-th power $[1 + \alpha(s + u) + su]^n$, where $\alpha = 1 - 2q$. Every entry is a multinomial sum weighted by powers of α; for integer α the matrix is automatically integer-valued. The edge entries are $I(0, k) = \binom{n}{k}\alpha^k$, and at $\alpha = 1$ the full matrix is $I(m, k) = \binom{n}{m}\binom{n}{k}$.

1 The Construction

The continuous Hermite matching matrix arises from Step 3 of the general construction: project the Hermite *functions* $\Phi_m = \widehat{H}_m\, e^{-t^2/2}$ back onto the Hermite polynomial basis $\{H_k\}$ in $L^2(e^{-t^2}\,dt)$. The projection coefficients involve the integral $\int H_m\, H_k\, e^{-3t^2/2}\,dt$, which is the expectation $\mathbb{E}[H_m(X)\, H_k(X)]$ for $X \sim N(0, 1/3)$. The factor $e^{-3t^2/2}$ comes from $e^{-t^2/2}$ (from Φ_m) times e^{-t^2} (from the weight).

We now perform the same construction in the discrete world. The Krawtchouk polynomials are the discrete analogues of the Hermite polynomials: they are orthogonal with respect to a binomial weight, just as the Hermite polynomials are orthogonal with respect to a Gaussian. And as the Gaussian is the continuous limit of the binomial distribution, everything we build here converges to the continuous Hermite theory as $n \to \infty$.

1.1 Krawtchouk polynomials

The *Krawtchouk polynomials* $K_j(x; n)$ for $p = \frac{1}{2}$ are defined on $\{0, 1, \ldots, n\}$ by the explicit sum

$$K_j(x; n) = \sum_{r=0}^{j} (-1)^r \binom{x}{r}\binom{n-x}{j-r}, \qquad 0 \le j \le n, \tag{1}$$

with generating function

$$\sum_{j=0}^{n} K_j(x; n)\, z^j = (1 - z)^x\, (1 + z)^{n-x}. \tag{2}$$

They satisfy the orthogonality relation

$$\sum_{x=0}^{n} K_j(x;n)\,K_k(x;n)\,\frac{\binom{n}{x}}{2^n} = \binom{n}{k}\,\delta_{jk}. \tag{3}$$

The Krawtchouk polynomials diagonalise the Hamming scheme: $K_j(x;n)$ is the character of the j-th stratum of the association scheme on $\{0,1\}^n$. The matrix $(K_j(k;n))_{j,k=0}^{n}$ is, up to normalisation, the Walsh–Hadamard transform — the Fourier transform of $(\mathbb{Z}/2\mathbb{Z})^n$, which is the binary analogue of the DFT.

1.2 The discrete matching matrix

Definition 1.1 (Krawtchouk matching matrix)**.** Fix $n \geq 1$ and a parameter $q \in [0,1]$ (or more generally $q \in \mathbb{Q}$). The *Krawtchouk matching matrix* $I(m,k,n,q)$ is defined by

$$I(m,k,n,q) = \sum_{x=0}^{n} K_m(x;n)\,K_k(x;n)\,\binom{n}{x}\,q^x\,(1-q)^{n-x} = \mathbb{E}\big[K_m(X;n)\,K_k(X;n)\big], \tag{4}$$

where $X \sim \mathrm{Bin}(n,q)$.

At $q = \frac{1}{2}$ this reduces to the orthogonality relation (3): the matrix is diagonal with entries $\binom{n}{k}$. For $q \neq \frac{1}{2}$ the off-diagonal entries are nonzero, and the matrix records how the Krawtchouk orthogonality structure "sees" a binomial weight at a different parameter.

2 The Generating Function

Theorem 2.1 (Bivariate generating function)**.** *Set $\alpha = 1-2q$. Then*

$$F(s,u) \;=\; \sum_{m,k=0}^{n} I(m,k,n,q)\,s^m\,u^k \;=\; \big[\,1+\alpha(s+u)+su\,\big]^n. \tag{5}$$

Proof. Using the generating function (2):

$$\begin{aligned}
F(s,u) &= \sum_{x=0}^{n}\binom{n}{x}\,q^x\,(1-q)^{n-x}\sum_{m}K_m(x;n)\,s^m\sum_{k}K_k(x;n)\,u^k \\
&= \sum_{x=0}^{n}\binom{n}{x}\,q^x\,(1-q)^{n-x}\,(1-s)^x(1+s)^{n-x}\,(1-u)^x(1+u)^{n-x} \\
&= \sum_{x=0}^{n}\binom{n}{x}\big[q(1-s)(1-u)\big]^x\,\big[(1-q)(1+s)(1+u)\big]^{n-x} \\
&= \big[\,q(1-s)(1-u)+(1-q)(1+s)(1+u)\,\big]^n.
\end{aligned}$$

Expanding the bracket:

$$q(1-s-u+su)+(1-q)(1+s+u+su) = 1+(1-2q)(s+u)+su. \qquad \square$$

Compare the two generating functions:

	Continuous (Hermite)	Discrete (Krawtchouk)
GF	$\sqrt{2\pi/3}\ \exp\!\left(-\frac{s^2-4su+u^2}{3}\right)$	$[1+\alpha(s+u)+su]^n$
Shape	exp of quadratic form	n-th power of affine–bilinear
Parameter	$\sigma^2 = 1/3$ (fixed)	$\alpha = 1-2q$ (free)

The continuous generating function is an *exponential* of a quadratic form; the discrete is an n-th *power* of an affine–bilinear form. They are connected by the usual limit: scaling $s \to s/\sqrt{n}$, $u \to u/\sqrt{n}$, $\alpha \to \alpha/\sqrt{n}$ and sending $n \to \infty$, the n-th power converges to the exponential.

3 Coefficient Extraction

Theorem 3.1 (Multinomial formula)**.** *The entry $I(m,k,n,q)$ is given by*

$$I(m,k,n,q) = \sum_c \frac{n!}{(n-m-k+c)!\,(m-c)!\,(k-c)!\,c!}\,\alpha^{\,m+k-2c}, \tag{6}$$

where c ranges over $\max(0,\, m+k-n) \le c \le \min(m,k)$. Each term is a multinomial coefficient times a power of α.

Proof. We expand $[1+\alpha(s+u)+su]^n$ by the multinomial theorem with three terms: pick a copies of 1, b copies of $\alpha(s+u)$, and c copies of su. Then $a+b+c=n$, and $(s+u)^b = \sum_j \binom{b}{j} s^j u^{b-j}$. The monomial $s^m u^k$ requires $j+c=m$ and $b-j+c=k$, giving $j=m-c$, $b=m+k-2c$, and $a=n-m-k+c$. The multinomial coefficient $\frac{n!}{a!\,b!\,c!}$ times the binomial $\binom{b}{j} = \binom{b}{m-c}$ simplifies to $\frac{n!}{a!\,(m-c)!\,(k-c)!\,c!}$. □

Corollary 3.2 (Automatic integrality)**.** *When $\alpha \in \mathbb{Z}$, every entry $I(m,k,n,q) \in \mathbb{Z}$. No rescaling is required.*

This is a striking contrast with the continuous case, where integrality required rescaling by $3^{(m+k)/2}$.

4 Basic Properties

Proposition 4.1 (Symmetries)**.** *The Krawtchouk matching matrix has the following symmetries.*

1. ***Transpose symmetry:*** $I(m,k,n,q) = I(k,m,n,q)$.
2. ***Complement symmetry:*** $I(m,k,n,q) = \alpha^n\, I(n{-}m,\, n{-}k,\, n,\, q)$.
3. ***Parameter reflection:*** $I(m,k,n,1{-}q) = (-1)^{m+k}\, I(m,k,n,q)$.
4. ***No parity vanishing:*** $I(m,k,n,q) \neq 0$ *in general when* $m+k$ *is odd (for* $\alpha \neq 0$*).*

Proof. (1) The generating function is symmetric in (s,u).

(2) From the Krawtchouk symmetry $K_j(n{-}x;n) = (-1)^j K_j(x;n)$: replacing x by $n{-}x$ in the sum and using $(1{-}q)^x q^{n-x} = \alpha^{-n}{\cdot}q^{n-x}(1{-}q)^x$ gives $I(n{-}m,n{-}k) = (-1)^{m+k}(-1)^{m+k}\alpha^{-n}I(m,k)$ $= \alpha^{-n}I(m,k)$.

(3) Replacing q by $1{-}q$ sends α to $-\alpha$; from (6), each term picks up $(-1)^{m+k-2c} = (-1)^{m+k}$.

(4) The generating function $[1+\alpha(s+u)+su]^n$ has monomials of all total degrees from 0 to $2n$, including odd degrees. □

4.1 Edge entries

Proposition 4.2 (Edge entries)**.** $I(0,k,n,q) = \binom{n}{k}\,\alpha^k$.

Proof. When $m=0$ the only term in (6) is $c=0$, giving $\frac{n!}{(n-k)!\,0!\,k!\,0!}\,\alpha^k = \binom{n}{k}\,\alpha^k$. Alternatively: $[s^0u^k]\,[1+\alpha(s+u)+su]^n = [u^k]\,(1+\alpha u)^n = \binom{n}{k}\,\alpha^k$. □

4.2 Special values of α

Proposition 4.3 (Factorisation at $\alpha = \pm 1$)**.**

1. $\alpha = 1$ *(i.e.* $q=0$*):* $F = [(1+s)(1+u)]^n$*, so* $I(m,k,n,0) = \binom{n}{m}\binom{n}{k}$.
2. $\alpha = -1$ *(i.e.* $q=1$*):* $F = [(1-s)(1-u)]^n$*, so* $I(m,k,n,1) = (-1)^{m+k}\binom{n}{m}\binom{n}{k}$.
3. $\alpha = 0$ *(i.e.* $q=\frac{1}{2}$*):* $F = (1+su)^n$*, so* $I(m,k) = \binom{n}{k}\,\delta_{mk}$ *(orthogonality).*

5 The Diagonal

Theorem 5.1 (Diagonal entries)**.** *The diagonal entry $I(k,k,n,q)$ is given by*

$$I(k,k,n,q) = \sum_{c=\max(0,2k-n)}^{k} \frac{n!}{(n-2k+c)!\,(k-c)!^2\,c!}\,\alpha^{2(k-c)}. \tag{7}$$

At $\alpha = 1$: $I(k,k,n,0) = \binom{n}{k}^2$.
At general integer α, the first several values are:

k	0	1	2	3	4	5	6	7	8
$\alpha = 1$	1	n	$\binom{n}{2}^2$						
$\alpha = 2,\ n = 8$	1	232	8092	66136	134470	66136	8092	232	1
$\alpha = 2,\ n = 6$	1	126	1935	4900	1935	126	1		

The diagonal at $\alpha = 1$ gives $\binom{n}{k}^2$, the central entries of the Delannoy triangle. The ordinary generating function of $\binom{n}{k}^2$ (as a polynomial in k for fixed n) is related to the Legendre polynomial P_n.
The diagonal at $\alpha = 2$ shows a unimodal sequence symmetric about $k = n/2$ (by the complement symmetry), but the entries do not have an obvious closed form. The central entry $I(n/2,\, n/2,\, n,\, q)$ (for n even) appears to grow as $(1+\alpha^2)^{n/2} \cdot$ polynomial.

6 Computed Values

The matrix $I(m,k,n,q)$ is symmetric in (m,k) and, for integer $\alpha = 1-2q$, has all entries in $\mathbb{Z}$.

$\alpha = 2,\ n = 6$

$m\backslash k$	0	1	2	3	4	5	6
0	1	12	60	160	240	192	64
1	12	126	540	1200	1440	864	192
2	60	540	1935	3480	3240	1440	240
3	160	1200	3480	4900	3480	1200	160
4	240	1440	3240	3480	1935	540	60
5	192	864	1440	1200	540	126	12
6	64	192	240	160	60	12	1

$\alpha = 2,\ n = 8$

$m\backslash k$	0	1	2	3	4	5	6	7	8
0	1	16	112	448	1120	1792	1792	1024	256
1	16	232	1456	5152	11200	15232	12544	5632	1024
2	112	1456	8092	24976	46480	52864	35392	12544	1792
3	448	5152	24976	66136	103600	97216	52864	15232	1792
4	1120	11200	46480	103600	134470	103600	46480	11200	1120
5	1792	15232	52864	97216	103600	66136	24976	5152	448
6	1792	12544	35392	52864	46480	24976	8092	1456	112
7	1024	5632	12544	15232	11200	5152	1456	232	16
8	256	1024	1792	1792	1120	448	112	16	1

$\alpha = 3$, $n = 6$

$m\backslash k$	0	1	2	3	4	5	6
0	1	18	135	540	1215	1458	729
1	18	276	1710	5400	8910	6804	1458
2	135	1710	8385	19620	21465	8910	1215
3	540	5400	19620	30800	19620	5400	540
4	1215	8910	21465	19620	8385	1710	135
5	1458	6804	8910	5400	1710	276	18
6	729	1458	1215	540	135	18	1

Several features are visible in the tables.

Persimmon symmetry. The matrix is symmetric about both diagonals: $I(m,k) = I(k,m)$ (transpose) and $I(m,k) = \alpha^n\, I(n-m, n-k)$ (complement). For $\alpha = 1$ both symmetries coincide and the matrix is a perfect square: $I(m,k) = \binom{n}{m}\binom{n}{k}$.

Unimodality. Each row (and column) appears to be unimodal — increasing to a maximum and then decreasing — when all entries are positive ($\alpha > 0$). The maximum of row m is at $k = \lfloor n/2 \rfloor$ or nearby.

Corners. The four corners are $I(0,0) = 1$, $I(0,n) = I(n,0) = \alpha^n$, $I(n,n) = 1$.

7 The Signed Case

$\alpha = -2$, $n = 6$

$m\backslash k$	0	1	2	3	4	5	6
0	1	−12	60	−160	240	−192	64
1	−12	126	−540	1200	−1440	864	−192
2	60	−540	1935	−3480	3240	−1440	240
3	−160	1200	−3480	4900	−3480	1200	−160
4	240	−1440	3240	−3480	1935	−540	60
5	−192	864	−1440	1200	−540	126	−12
6	64	−192	240	−160	60	−12	1

By Proposition 4.1(3), sending α to $-\alpha$ multiplies entry (m,k) by $(-1)^{m+k}$, so the $\alpha = -2$ matrix is the $\alpha = 2$ matrix with a checkerboard sign pattern. This is the discrete analogue of the sign structure in the continuous Hermite matrix, where D_{mk} alternates in sign along each row.

8 Rational α: Rescaling to Integers

For non-integer α the entries are rational. But rescaling by d^{m+k}, where d is the denominator of α, restores integrality — just as in the continuous case, where the rescaling $3^{(m+k)/2}$ absorbed the denominators from $e^{-3t^2/2}$.

Proposition 8.1 (Rational rescaling)**.** *If $\alpha = p/d$ with $p, d \in \mathbb{Z}$, $d \geq 1$, then*

$$d^{m+k}\, I(m,k,n,q) \;\in\; \mathbb{Z} \qquad \text{for all } m, k.$$

Proof. Each term in (6) contains $\alpha^{m+k-2c} = p^{m+k-2c}/d^{m+k-2c}$. Multiplying by d^{m+k} gives $d^{2c}\, p^{m+k-2c}$ times a multinomial coefficient, which is integral. □

The case $\alpha = 1/3$ (i.e. $q = 1/3$) deserves attention: it is the nearest discrete echo of the Hermite variance $\sigma^2 = 1/3$. The rescaled matrix $3^{m+k}\, I(m, k, 6, 1/3)$ is:

$\alpha = 1/3$, $n = 6$, rescaled by 3^{m+k}

$m\backslash k$	0	1	2	3	4	5	6
0	1	6	15	20	15	6	1
1	6	84	330	600	570	276	54
2	15	330	2385	6540	8385	5130	1215
3	20	600	6540	30800	58860	48600	14580
4	15	570	8385	58860	193185	240570	98415
5	6	276	5130	48600	240570	551124	354294
6	1	54	1215	14580	98415	354294	531441

Note the edge: the rescaled $I(0, k) = 3^k \cdot \binom{6}{k}/3^k = \binom{6}{k}$, recovering the binomial coefficients. The corner $I(6, 6) = 3^{12} \cdot I(6, 6, 6, 1/3) = 3^{12}/3^6 = 3^6 = 531441$. The entry $I(3, 3) = 30800$ is the same as in the $\alpha = 2$ table — a coincidence from the multinomial formula, since the diagonal value at $k = n/2$ depends on α^2, and both $\alpha = 2$ and $\alpha = 1/3$ give distinct values of α^2 (so in fact this is *not* a coincidence: the rescaled $\alpha = 1/3$ diagonal is simply different from the $\alpha = 2$ diagonal, and the agreement at $(3, 3)$ is the complement symmetry forcing $I(3, 3) = \alpha^6\, I(3, 3)$, hence $\alpha^6 = 1$, which fails for both — the agreement is that $I(3, 3, 6, q) = 4900$ for $\alpha = 2$ and the rescaled value is 30800 for $\alpha = 1/3$; these are different).

9 Structural Comparison

Feature	Continuous (Hermite)	Discrete (Krawtchouk)
Generating fn	$\sqrt{\frac{2\pi}{3}}\,\exp\!\left(-\frac{s^2-4su+u^2}{3}\right)$	$[1+\alpha(s+u)+su]^n$
Analytic shape	exp of quadratic	n-th power
Free parameter	none ($\sigma^2=1/3$)	$\alpha\in\mathbb{Z}$
Integrality	rescale by $3^{(m+k)/2}$	automatic
Parity	$D_{mk}=0$ for $m{+}k$ odd	no vanishing
Transpose sym	$D_{mk}=D_{km}$	$I(m,k)=I(k,m)$
Extra sym	—	complement: $I(m,k)=\alpha^n I(n{-}m,n{-}k)$
Edge	$D_{0,2n}=(-1)^n(2n)!/n!$	$I(0,k)=\binom{n}{k}\alpha^k$
At $\alpha=1$	—	$I(m,k)=\binom{n}{m}\binom{n}{k}$
Diagonal OGF	$\frac{1}{\sqrt{(1-2x)(1-6x)}}$	(open)
Combinatorics	signed bipartite matchings	multinomial coefficients

The continuous matrix has one parameter (the variance $1/3$, forced by the construction). The discrete matrix has a free integer parameter $\alpha = 1-2q$, and distinct values of α give genuinely different integer matrices. The continuous theory emerges as the $n\to\infty$ limit.

10 Open Questions

Conjecture 10.1 (Combinatorial interpretation)**.** *For integer $\alpha \geq 1$, the entry $I(m,k,n,q)$ counts something: weighted lattice paths, coloured set partitions, forests, or a discrete analogue of bipartite matchings. The multinomial formula* (6) *already gives a sum of positive terms (for $\alpha>0$), each being a multinomial coefficient times α^{m+k-2c}. This is a weighted count of ways to partition n labelled objects into four groups of sizes $(n{-}m{-}k{+}c,\ m{-}c,\ k{-}c,\ c)$, with α^{m+k-2c} colourings of the middle two groups. A bijective proof would illuminate the connection to bipartite matchings in the $n\to\infty$ limit.*

Conjecture 10.2 (Diagonal generating function)**.** *For fixed α and varying n, the sequence $I(k,k,n,q)$ (as a function of k for fixed n, or as a sequence in n for fixed $k>0$) should have a closed-form generating function involving algebraic functions or hypergeometric series, generalising the Hermite diagonal's $1/\sqrt{(1-2x)(1-6x)}$.*

Conjecture 10.3 (Scaling limit)**.** *Under the substitutions $m=\lfloor\mu\sqrt{n}\,\rfloor$, $k=\lfloor\kappa\sqrt{n}\,\rfloor$, $\alpha=\beta/\sqrt{n}$ with β chosen so that α^2 matches the Hermite parameter, the discrete matrix converges:*

$$\lim_{n\to\infty} I\big(\lfloor\mu\sqrt{n}\,\rfloor,\ \lfloor\kappa\sqrt{n}\,\rfloor,\ n,\ q\big)\ \sim\ D^{Hermite}_{\mu,\kappa}\ \cdot\ (\textit{normalisation}).$$

11 Proofs Available Upon Request

All entries in the tables were computed directly from the proved multinomial formula (Theorem 3.1) and independently verified against the direct Krawtchouk sum (4) for $n \leq 12$ and all $\alpha \in \{-3, -2, -1, 0, 1, 2, 3, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{2}\}$.